Agentic Video Understanding: A Survey

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***Abstract*—Video understanding is shifting from fixed video-language inference to agentic systems that decide what evidence to inspect, retain, verify, and act upon. This survey reviews video understanding agents: systems that use video as the primary source and solve understanding tasks through adaptive state construction and action selection. We first define the scope of video understanding tasks and formalize an agent loop over available video evidence, tools, observations, and intermediate states. We then organize the literature through a challenge-to-design taxonomy, linking context bottlenecks to hierarchical evidence memory, evidence sparsity to active acquisition, temporal causality to state and process tracking, and multimodal ambiguity to role-specialized coordination. We further review the state-space abstractions, learning paradigms, supervision signals, benchmarks and evaluations. Finally, we identify open directions toward agentic-native temporal modeling and video-native agents.**

***Index Terms*—video understanding agents, agentic video understanding.**

I. Introduction

Video understanding has evolved from recognition-centered modeling to open-ended reasoning over long, multimodal, and interactive videos. Early *video networks* treated video as a spatiotemporal signal and learned representations for action recognition, localization, captioning, and retrieval through temporal convolution, recurrent modeling, and space–time attention [1]–[4]. More recently, *Video Large Language Models (Video LLMs)* have reframed video as an input for language-based interaction, enabling models to answer questions, summarize events, localize moments, and reason over video content through visual tokens, captions, and timestamps [5]–[8]. This progression has greatly expanded the interface of video understanding, but it has not removed a deeper limitation: most systems still assume a largely passive inference pipeline, where frames or clips are sampled, encoded, placed into context, and decoded into an answer.

The weakness of this pipeline becomes apparent when the relevant evidence cannot be determined easily. In long videos, most frames are redundant, while the answer may depend on a brief event, a distant earlier state, or a cue from speech, sound, text, or interaction. In streaming settings, the problem is even harder because future evidence may not yet be observable, and an immediate answer may be premature. These conditions turn video understanding from a fixed encoding problem into an adaptive evidence-control problem: the system needs to decide where to look, what to store, which evidence to trust, whether more observation is needed, and when a response is justified.

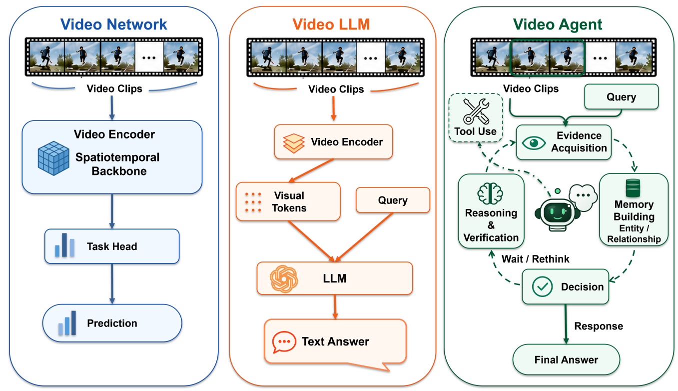

Fig. 1. From Video Networks to Video Agents: The Shift from Passive Video Processing to Adaptive Evidence Control.

Video agents emerge from this shift by introducing control into the video understanding loop. Rather than following a predetermined path from sampled frames to an answer, they can adaptively inspect video segments, invoke perception tools, update evidence states, verify hypotheses, and decide whether to continue observing or respond [9]–[14]. As illustrated in Fig. 1, this marks a shift from passive video processing to adaptive evidence control. A formal definition and the corresponding inclusion criteria are provided in Section II.

Existing surveys focus on video-language understanding, video foundation models, and Video LLMs [15]–[17]. Yet they do not systematically cover the emerging field of video agents. This survey addresses this gap by organizing agentic video understanding around the interaction between video-specific challenges and agentic control. This survey follows the structure below.

1) We define *video understanding agents* as systems that solve video tasks through adaptive state construction and action selection, thereby distinguishing them from conventional Video LLMs with fixed inference pipelines.
2) We develop a challenge-to-design taxonomy that explains why agentic mechanisms are needed for video understanding: context bottlenecks motivate hierarchical evidence memory, evidence sparsity motivates active evidence acquisition, temporal causality motivates state and process tracking, and multimodal ambiguity motivates role-specialized coordination.
3) We introduce a state-space view of agentic video understanding, tracing how video is represented as a bag of

frames, a temporal sequence, a graph of entities, or an evolving world state.
4) We further examine how video agent behaviors are specified or learned, covering inference-time control, supervised imitation, and reinforcement learning, together with the trajectory, grounding, and reward signals that support them. The full discussion is provided in Section I and Section.II of the Appendix
5) We review benchmarks, evaluations, and future directions for the field.

II. Scope and Definitions

A. Video Understanding Tasks

We use *video understanding* to refer to tasks in which video is the primary input source and the system interprets frames, temporal structure, auditory signals, video-associated text, or interaction cues to produce an answer, description, localization, decision, or structured output. The defining requirement is that the output depends on information contained in the video, whether this information is captured from individual frames, short clips, temporal order, multimodal context, entity relations, state changes, or streaming interaction. Thus, the task cannot be reduced to processing isolated images or external text alone. This scope covers video question answering, captioning, temporal grounding, moment retrieval, summarization, anomaly detection, referring video object segmentation, egocentric reasoning, streaming video understanding, and other video-conditioned tasks.

B. Video Understanding Agents

To distinguish video understanding agents from fixed-inference Video LLMs, we formalize video understanding as an adaptive decision process over video evidence. Let

$$V = \{x_\tau\}_{\tau=1}^{T}$$

denote a temporally ordered video, where x_τ denotes the multimodal observation available at time τ, including visual content and any video-derived auxiliary signals such as audio, text, metadata. Let q denote the task instruction, $\mathcal{T}$ the available tools, and s_k the intermediate evidence state at decision step k. We use $\bar{V}_k$ to denote the video evidence available at that step: $\bar{V}_k = V$ for offline video, while in streaming settings it contains only the observed prefix.

We define a *video understanding agent* as a system that uses video as its primary source and solves the task through adaptive evidence-state construction and action selection:

$$\begin{aligned} a_k &\sim \pi_\theta(a \mid q, s_{k-1}, \bar{V}_k, \mathcal{T}), && (1)\\ o_k &= \mathcal{O}(\bar{V}_k, a_k, \mathcal{T}), && (2)\\ s_k &= \mathcal{U}(s_{k-1}, o_k, a_k, q). && (3) \end{aligned}$$

Here, an action may change what evidence is accessed, how the state is updated, whether evidence is verified, or when the agent stops to produce an output. The resulting observation o_k is incorporated into the updated evidence state s_k. If $K = \min\{k : a_k = \textsc{Stop}\}$, the final output is

$$y = \mathcal{D}_\theta(q, s_K).$$

A system satisfies this definition when it maintains an explicit evidence state and adaptively selects at least one action that changes subsequent evidence access, state update, tool use, interaction, or termination. A standard Video LLM with fixed sampling and one-pass decoding is therefore excluded, whereas adaptive retrieval, memory, multi-agent, streaming, and proactive video systems are included.

C. Survey Corpus and Inclusion Criteria

We construct the survey corpus primarily from recent papers in major machine learning and computer vision venues, including ICML, ICLR, NeurIPS, CVPR, ICCV, and ECCV, together with highly relevant recent preprints when they introduce agentic video-understanding systems not yet represented in archival venues. Since video understanding agents have emerged mainly in the past few years, we focus on papers published or released between 2024 and 2026, while retaining earlier foundational benchmarks, datasets, and video-understanding methods when they are necessary for task or evaluation context.

A work is included in the core corpus when it satisfies three criteria:

- video serves as a primary input source;
- the task falls within the scope of video understanding defined above;
- the method exhibits agentic control as defined in the Video Understanding Agents subsection.

Under this criterion, the survey covers offline and online settings, short and long videos, single-video and multi-video tasks, and emerging interactive or streaming agentic tasks.

Adjacent areas such as embodied or egocentric agents that learn from video demonstrations [18]–[20], computer-use agents trained from screen recordings [21]–[23], and video generation or editing agents [24], [25] are considered only when they clarify mechanisms relevant to video understanding, rather than as primary targets of the survey.

III. Challenge-to-Design Taxonomy

The central claim of our taxonomy is that video agents should be organized not merely by the mechanisms they use, but by the video-specific challenge that makes an agentic mechanism necessary. Instead of asking whether a system contains memory, tools, state representations, or multiple agents, we ask what bottleneck in video understanding these designs are meant to address. General-purpose agents also use memory, tools, state, and collaboration, but video gives these designs sharper roles: memory retains evidence under long temporal and token budgets, search acquires sparse query-relevant evidence, state tracking models temporal change and causal continuity, and collaboration coordinates heterogeneous modalities and perceptual roles.

The complete challenge-to-design taxonomy is provided in Fig. S1 of the Appendix.

A. Context Bottleneck: Why Video Agent Needs Hierarchical Memory

The context bottleneck is the most basic pressure in video understanding and becomes especially severe in long-form, egocentric, multi-source, or streaming settings. A passive Video LLM typically compresses long visual histories into a fixed context window, where frames, captions, transcripts, and tool outputs compete for limited token and compute budgets. Even when a long-context model can ingest more frames, the information is not equally useful: most frames are redundant, while a small object, a brief utterance, or a single state change may matter later. This makes flat prompting and flat summarization brittle. They compress video before the query-relevant structure is known, and often lose the key information.

Hierarchical evidence memory addresses this bottleneck by making video evidence addressable at multiple temporal and semantic levels. WorldMM [26] separates episodic, semantic, and visual memories, allowing a retrieval agent to choose the memory source and scale that fit the query. VideoARM [27] uses an observe-think-act-memorize loop to build hierarchical multimodal memory rather than repeatedly reprocessing the same evidence. HAVEN [28] adds audiovisual entity cohesion, preserving links between people, objects, sounds, and temporal segments. AVI [29] builds an entity-graph video knowledge base with retrieve-perceive-review, while Embodied VideoAgent [18] extends memory with egocentric sensors such as depth and pose.

The key difference from generic agent memory is provenance and temporal granularity. A video memory item should not merely be a sentence in a scratchpad, it should point back to frames, intervals, modalities, entities, even the uncertainty. Otherwise memory will become a second source of hallucination. Strong video agents therefore need memories that are compact enough for reasoning but still grounded enough to recover the original evidence.

Recent systems instantiate the same memory argument under more demanding input regimes. Flash-VStream [30], StreamChat [31], OmniLive [32], and ProVideLLM [33] use cache or memory structures to avoid repeatedly processing the full visual history, while StreamMeCo [34] studies how to compress long-term agent memory without losing retrieval quality. Their contribution is to make memory operations explicit under severe compression and retrieval constraints. Comparing with observation behavior, this dimension more concerns evidence retention and the next dimension concerns how the agent decides which evidence should be observed in the first place.

B. Evidence Sparsity: Why Video Agent Needs Active Evidence Acquisition

Many video questions are needle-in-a-haystack problems. The relevant event may be a two-second gesture in an hour-long stream, a single spoken phrase in a lecture, a rare anomaly in surveillance footage, or one object interaction among many similar distractors. A general Video LLM with fixed frame sampling can be accurate only when the sample happens to include the answer-critical evidence. The core challenge is not how to store all observed content, but how to decide what additional evidence should be observed next. Agentic systems improve this by turning perception into an adaptive acquisition process: observe partial evidence, infer what is still missing, call a tool or inspect a segment, and repeat.

Early video agents make the decision about the next observation explicit. VideoAgent [9] uses an LLM controller to plan subgoals and retrieve visual evidence relevant to the task. VCA [35] explores candidate video segments according to curiosity signals, while VideoAgent2 [12] uses uncertainty in the reasoning process to decide when incomplete observations require further inspection. VideoSeek [36] and LensWalk [37] extend this loop by letting the agent choose among tools, temporal scopes, and sampling densities; EVA [38] makes the seek, plan, and reflect process trainable. Flow4Agent [39] is a related efficiency case, using motion priors to suppress redundant temporal and spatial information before reasoning.

This acquisition problem becomes broader when evidence is distributed across tools, modalities, or sources rather than only across time. DVD [40] treats long video as a searchable database with tool calls at different levels of granularity. VideoExplorer [41] decomposes questions, grounds relevant moments, and performs perception guided by the task, while OmAgent [10] coordinates retrieval, memory access, and direct video inspection through task decomposition. MAGNET [42] extends acquisition to audiovisual haystacks across multiple videos, where sparse evidence may lie in another modality or another video.

Online settings further stress active acquisition because relevant evidence may lie in past observations, current frames, or future events. StreamAgent [43] casts this as anticipatory evidence selection over temporal intervals and spatial regions, while AViLA [13] studies cases where the user query and the supporting visual evidence are not synchronized. Thus, active evidence acquisition is the agentic control of where, when, and through which tool or modality missing video evidence should be obtained.

C. Temporal Causality: Why Video Agent Needs State Tracking

Video is not only a collection of visual observations but an ordered record of change. Many questions cannot be answered by retrieving a single salient frame or segment. They ask what happened before another event, whether an object changed state, why a person acted, how a scene evolved, or which entity persisted across cuts and viewpoints. This is where passive video-language inference often fails: it can recognize local content but lose the chain of state transitions that makes an answer causal. This challenge is distinct from memory and active acquisition. Memory concerns whether past evidence remains accessible, and active acquisition concerns where additional evidence should be found, whereas state tracking concerns how observed evidence is linked into evolving entities, relations, actions, and event chains.

State tracking makes temporal structure explicit. It can appear as an entity timeline, a scene graph that is updated over time, a storyline, a hypothesis ledger, or a memory in which each entry records what changed and when. SVAgent [44] organizes video understanding around a storyline that guides later reasoning. HAVEN [28] tracks cohesion among audio-visual entities across long videos, while WorldMM [26] and VideoARM [27] use memory structures that allow the agent to retrieve events at appropriate temporal scales.

A memory can store a caption such as "the person enters the kitchen," but a state tracker should also represent whether the person later leaves, what object they carry, and which later event depends on that earlier state. This distinction matters because many agentic gains come not simply from remembering more evidence, but from preserving evidence as a temporally organized structure.

Long horizon and online settings make this requirement more explicit. EGAgent [45] represents egocentric video with entity scene graphs and temporal intervals, giving the agent a substrate to track people, objects, locations and events over long horizons. ThinkStream [14] and StreamEQA [46] evaluate related forms of temporal state reasoning: the answer depends on whether the agent is using past evidence, interpreting the current state, or anticipating possible next steps.

D. Multimodal Ambiguity: Why Video Agent Needs Role-Specialized Coordination

Video evidence is multimodal and often ambiguous. The visual stream may show an action without revealing intent; speech may explain an event that is off camera; audio may identify a sound source before it appears visually; Optical Character Recognition (OCR) may disambiguate a location; and motion may contradict a static frame caption. A single agent can process these signals, but a single reasoning stream may overcommit to the most fluent modality and ignore weak contradictory evidence.

The key design response is role-specialized multimodal coordination, not merely multi-agent collaborationS. The relevant question is how a system separates and reconciles heterogeneous video evidence across modalities, rather than whether it simply contains multiple agents. Multi-agent collaboration is one common implementation, but module-specialized architectures can serve the same purpose when they assign distinct perceptual or reasoning roles.

Module-specialized multimodal systems are therefore relevant here because they separate functions that a single monolithic model tends to blur. OmniLive [32] separates perception, memory, and reasoning modules; AViLA [13] coordinates memory retention, evidence identification, and triggering; and ViSpeak [47] studies visual instructions such as gesture-based wake-up or interruption. These examples are best read as role-specialized perception and coordination mechanisms adjacent to, but not reducible to, multi-agent collaboration.

Multi-agent collaboration is useful when video ambiguity benefits from role specialization. One agent can plan the search, another can ground temporal evidence, another can inspect visual details, another can check audio or transcript cues, and another can synthesize or critique the answer. VideoMind [48] implements planner, grounder, verifier, and answerer roles through a Chain-of-LoRA mechanism. LVAgent [49] frames long-video understanding as multi-round collaboration among multimodal large language model (MLLM) agents. Symphony [50] separates grounding, visual perception, planning, and reflection in a cognitively inspired architecture. A4VL [51] forms a perception-action alliance in which vision-language model (VLM) agents extract clues, align them to blocks, cross-review one another, and continue until consensus.

Other systems stress deliberation, self-correction, or policy coordination. ReAgent-V [52] uses reward-driven critic feedback and can generate training data for SFT, Direct Preference Optimization (DPO), and Group Relative Policy Optimization (GRPO). VideoChat-M1 [53] uses collaborative policy planning and optimizes the collaboration with multi-agent reinforcement learning. Takusen [54] separates streaming dense-captioning decisions between an Oracle agent that detects boundaries and a Listener agent that captions content. MAGNET [42] uses multi-agent reasoning over audio-visual haystacks. These systems are most central to this dimension when collaboration improves the use of heterogeneous video evidence, rather than merely increasing inference-time compute or adding generic deliberation. Future evaluations should therefore measure not only final accuracy, but also whether role specialization improves evidence coverage, cross-modal correction, agreement diversity, and groundedness.

IV. Paradigm: State Space

Having linked video-specific challenges to agentic design choices, we next examine the operative video states on which agents reason and act. We organize existing methods into four dominant paradigms: *Video as a Bag of Frames*, *Video as a Temporal Sequence*, *Video as a Graph of Entities*, and *Video as an Evolving World State*. These paradigms differ in whether video evidence is organized as discrete visual units, temporally related observations, persistent relational structures, or causally updated states as the video unfolds.

A. Paradigm I: Video as a Bag of Frames

In this paradigm, a video is not primarily represented as a continuous temporal process or as a persistent long-term memory, but is instead abstracted into a set of discrete visual evidence units, such as frames, keyframes, clips, shots, or candidate video segments. Methods in this paradigm may retain local temporal cues or basic ordering information, but they do not explicitly organize the evidence around temporal transitions, event evolution, or long-term state dependencies. Once a frame, clip, or segment is selected, it functions mainly as a piece of evidence for recognition, grounding, or answer generation. The resulting video state is therefore closer to a collection of query-relevant visual observations than to an explicit model of how events or states evolve over time.

Different methods instantiate this state space at different granularities. Some represent the video through sparse frames or keyframes. EVA [38] treats perception as the construction of a compact set of visual observations before reasoning, while APPO [55] optimizes attention-guided visual evidence allocation over sampled frames or tokens. Refer-Agent [56] adopts a similar state assumption in referring video object segmentation: the video is first reduced to query-relevant keyframes, on which object grounding is performed before mask propagation.

Another group uses coarser evidence units, such as chunks, blocks, or candidate segments. A4VL [51] partitions a long video into visual blocks and reasons over the subset most relevant to the query. LVAgent [49] similarly represents long videos through retrieved chunks that provide candidate evidence for multi-agent collaboration. SALOVA [57] uses segment-level representations and routing for long-form video analysis, while Flow4Agent [39] uses optical-flow priors to form motion-aware evidence units. Across these granularities, temporal cues may guide evidence construction, but the operative state remains a collection of selected visual units rather than an explicit representation of temporal evolution.

Overall, Video as a Bag of Frames offers a compact and efficient state-space abstraction that is compatible with existing VLMs and directly grounded in visual evidence. However, without explicit modeling of temporal order, entity persistence, or state transitions, it remains limited for reasoning about how events unfold and the video world changes over time.

B. Paradigm II: Video as a Sequence of Frames

Paradigm II represents video as an ordered sequence of observations. Compared with the bag-of-frames abstraction, the video state is determined not only by which frames, clips, or segments are retained, but also by their positions and relations along the timeline. Methods in this paradigm make temporal structure explicit, so a segment is interpreted through its position in an unfolding process rather than as isolated evidence.

Many long-video agents instantiate this paradigm through temporally organized evidence spaces. VideoAgent [9] and VideoAgent2 [12] search over temporally localized segments and progressively refine evidence along the video timeline. VCA [35] organizes long videos into hierarchical temporal segments, while LensWalk [37] treats observation as a multi-scale path through the video. VideoChat-A1 [58] further decomposes videos into shot and subshot chains, allowing reasoning to follow their temporal organization. In these methods, the relations among selected segments, rather than segment relevance alone, contribute to the operative video state.

This sequential abstraction is also central to procedure and trajectory modeling. Mobile-Agent-V [23] and VideoAgentTrek [21] model screen recordings as ordered GUI state transitions. RoomTour3D [19] represents room-tour videos as spatiotemporal trajectories for embodied navigation. AVT [59] similarly evaluates clips within a coherent sequence, such that their meaning depends partly on preceding and subsequent observations.

In conclusion, this paradigm captures video as an unfolding process whose observations are organized by temporal relations. Unlike the Video as a Graph of Entities paradigm introduced next, its defining representation remains an ordered temporal sequence rather than a persistent entity-centric relational structure.

C. Paradigm III: Video as a Graph of Entities

Paradigm III organizes video as a relational state rather than only as an ordered sequence of observations. Here, an "**entity**" denotes any reusable state unit that can persist, recur, or support later reasoning across the video. Likewise, a "**graph**" need not be implemented as a formal graph data structure, but it must make associations across frames, segments, modalities, or reasoning steps explicit. These associations may represent identity, co-reference, temporal or spatial relations, multimodal alignment, or evidential support. Unlike Paradigm II, temporal order may still be preserved, but it is no longer the sole structure through which video evidence is organized.

A first line of work builds structured evidence archives from long videos. DrVideo [11] converts videos into document-style memory, where segment descriptions can be retrieved and supplemented with visual details. OmAgent [10] constructs a multimodal memory from scene captions, automatic speech recognition (ASR) transcripts, face recognition, timestamps, and vector databases. Video-RAG [60] and AdaVideoRAG [61] further organize OCR, ASR, object texts, visual embeddings, and knowledge-graph-like records into retrievable evidence states. MR.Video [62] follows a MapReduce formulation, aggregating scene-level captions while resolving references to persons and objects.

A second line makes entities and relations more central. VideoAgent [63] maintains temporal and object memory for long-video understanding. GraphVideoAgent [64] explicitly represents long videos with entity-relation graphs. EGAgent [45] extends this idea to days- or weeks-long egocentric videos through entity scene graphs over people, places, objects, and events. HAVEN [28] uses audiovisual entity cohesion to connect visual, audio, and textual evidence around persistent entities. WorldMM [26] further separates episodic, semantic, and visual memory, producing a multi-scale state that links events, long-term relations, and visual details.

Relational video states may also be constructed dynamically around the query. VideoARM [27] organizes video intervals, multimodal clues, and tool outputs in a hierarchical memory, while VideoChat-M1 [53] shares discovered video clues and intermediate results across agents to guide further exploration.

In summary, this paradigm supports long-range association, multi-hop retrieval, entity persistence, and multimodal evidence fusion. Its main risk is representational distortion: if captions omit details, graphs link entities incorrectly, or memory states compress away relevant evidence, subsequent reasoning may be structurally organized but factually misgrounded.

D. Paradigm IV: Video as an Evolving World State

Paradigm IV represents video as a causally updated state of an unfolding situation. Unlike Paradigm III, whose defining structure lies in explicit associations among evidence units, Paradigm IV is defined by causal state maintenance: the representation must remain valid as new observations arrive while future frames remain unavailable. The state is therefore time-indexed, partial, and incrementally revisable. It captures what has been observed, the current interpretation of the video, and how incoming evidence changes that interpretation; it may also track uncertainty, missing evidence, or response readiness.

This distinction is most visible in streaming and interactive settings. AViLA [13] addresses the asynchronous arrival of user queries and supporting visual evidence, requiring the system to distinguish observed from not-yet-observed information. ThinkStream [14] formulates streaming understanding as an incremental Watch–Think–Speak process, in which each incoming chunk may revise the current state before a response is produced. R3-Streaming [65] further connects state updates with readiness estimation, forgetting, and adaptive fast-slow routing. Streaming Harness [66], ViSpeak [47], and OmniLive [32] extend this setting to open-ended interaction and long-term audiovisual dialogue, where the system must jointly maintain evolving video and conversational context.

Other works emphasize prediction, risk, and embodied scene updates. StreamAgent [43] maintains a state that supports anticipation of possible future events rather than only summarizing past observations. Embodied VideoAgent [18] updates object locations and scene states from egocentric video and embodied sensing. PANDA [67] treats surveillance video as an evolving risk context that combines scene information, anomaly rules, and uncertain observations. EgoAgent [20] represents egocentric interaction through interleaved state-action tokens, linking observation, prediction, and action. Taken together, this paradigm is defined not by online memory alone, but by whether the video representation remains causally valid and continuously revisable as the video unfolds.

V. Benchmarks and Evaluation

Table II summarizes benchmarks relevant to agentic video understanding. We distinguish two complementary roles. *Capability-oriented benchmarks* measure skills required by video agents, including temporal grounding, long-range evidence integration, multimodal reasoning, and procedural understanding. *Agent-oriented benchmarks* evaluate whether agents control evidence access and response decisions correctly. Because some benchmarks assess both, so we categorize them by their primary evaluation focus.

Capability-oriented evaluation has expanded from short-video question answering (QA), temporal localization, and procedure understanding to long-context and diagnostic reasoning. Early benchmarks such as Charades-STA [89], ActivityNet-QA [90], YouCook2 [91] and NExT-QA [92] established temporal grounding, open-ended QA, instructional-video understanding, and causal reasoning. More recent benchmarks, including EgoSchema [93], MVBench [94], LongVideoBench [95], LVBench [96], MLVU [97], TempCompass [98], and Video-MME [99], extend evaluation to long-range egocentric reasoning, multimodal diagnosis, and increasingly long video contexts. These benchmarks measure important capabilities required by video agents, but typically provide the complete video before prediction.

Agent-oriented evaluation adds constraints on when evidence becomes available and what the system must decide or do. StreamingBench [100] bridges conventional and agent-oriented evaluation through timestamped video prefixes, sequential QA, and proactive-output simulation. OVO-Bench [101], OmniMMI [102], and OmniPro [103] further require causal video access and decisions about when to respond. Ego2Web [104] and VideoDR [105] extend video-conditioned evaluation to web execution and external information retrieval. These benchmarks shift the question from only what a model can answer to when the evidence is available and what the system should do next.

Across these benchmarks, evaluation varies along three axes: the *evidence horizon*, from short clips to distributed long-video evidence; the *access regime*, from complete-video input to causal observation; and the *action scope*, from answer generation to proactive response and external execution. However, most benchmarks still emphasize final-task success. More complete evaluation should assess not only final-task success, but also the quality, timing, cost, and reliability of the agent's evidence-use process.

VI. Open Challenges and Future Directions

A. Agentic-Native Temporal Modeling

Existing video models encode time through temporal convolution, recurrent states, space–time attention, or timestamp-aware tokens and memory structures [1]–[8]. For video agents, temporal information can play a more active role when it is exposed as an operable state rather than encoded only as part of the input. The agent should be able to query what has been observed, trace the evidence supporting its current beliefs, revise uncertain states, and use this information to guide later verification, response, or action.

Current video agents distribute these functions across search, memory, verification, and streaming-control modules [9], [11]–[14], [36], [37], [44], [48], [100], [102], [117]. A key direction is to unify them through a temporal-state interface that supports explicit grounding, reading, updating, and stopping operations. Such an interface should connect temporally anchored evidence and uncertainty to decisions about whether to observe, retrieve, verify, wait, respond, or act. Evaluation should test whether agents preserve correct temporal beliefs, update them as evidence arrives, select the right moments, and avoid premature or unsupported decisions.

B. Video-Native Agent Modeling

Current video agents remain largely LLM-centric. Video content is often converted into textual summaries, retrieved evidence, structured memories, or multimodal tokens, after which most reasoning occurs in a language-dominated

TABLE I
TAXONOMIC OVERVIEW OF VIDEO UNDERSTANDING AGENT METHODS. METHODS ARE ORDERED BY PUBLICATION YEAR. ADDITIONAL METHODS ARE REPORTED IN TABLE S1 IN THE APPENDIX.

Method	**Year**	**Challenge**	**Paradigm**	**Learning Paradigms**			**Data and Supervision**		
				Training-Free Control	Supervised Fine-Tuning	Reinforcement Learning	Trajectory Supervision	Grounding Supervision	Reward Signals
AVT [59]	2024	Temporal Causality	Paradigm II	★	☆	☆	☆	☆	☆
DoraemonGPT [68]	2024	Temporal Causality	Paradigm III	★	☆	☆	☆	☆	☆
IXC2.5-OL [32]	2024	Multimodal Ambiguity	Paradigm IV	★	☆	☆	☆	☆	☆
OmAgent [10]	2024	Evidence Sparsity	Paradigm III	★	☆	☆	☆	☆	☆
SALOVA [57]	2024	Evidence Sparsity	Paradigm I	☆	☆	☆	☆	☆	☆
SlowFocus [69]	2024	Evidence Sparsity	Paradigm II	☆	☆	☆	☆	☆	☆
Video-of-Thought [70]	2024	Temporal Causality	Paradigm III	★	★	☆	☆	★	☆
VideoAgent (Fan et al.) [63]	2024	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
VideoAgent (Wang et al.) [9]	2024	Evidence Sparsity	Paradigm II	★	☆	☆	☆	☆	☆
VideoStreaming [71]	2024	Context Bottleneck	Paradigm IV	☆	★	☆	☆	★	☆
AdaVideoRAG [61]	2025	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
AoTD [72]	2025	Evidence Sparsity	Paradigm II	☆	★	☆	★	☆	☆
AVI [29]	2025	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
AViLA [13]	2025	Temporal Causality	Paradigm IV	★	☆	☆	☆	★	☆
Commonsense Video QA [73]	2025	Evidence Sparsity	Paradigm II	★	☆	☆	☆	★	☆
DrVideo [11]	2025	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
DVD [40]	2025	Evidence Sparsity	Paradigm III	★	☆	☆	☆	☆	☆
EgoAgent [20]	2025	Temporal Causality	Paradigm IV	☆	★	☆	★	☆	☆
Embodied VideoAgent [18]	2025	Temporal Causality	Paradigm IV	★	☆	☆	☆	☆	☆
Eyes Wide Open [74]	2025	Temporal Causality	Paradigm IV	☆	★	☆	★	☆	☆
Flash-VStream [30]	2025	Context Bottleneck	Paradigm III	☆	☆	☆	☆	☆	☆
Flow4Agent [39]	2025	Evidence Sparsity	Paradigm I	☆	☆	☆	☆	☆	☆
FrameThinker [75]	2025	Evidence Sparsity	Paradigm I	☆	★	★	★	☆	★
GraphVideoAgent [64]	2025	Temporal Causality	Paradigm III	★	☆	☆	☆	☆	☆
LION-FS [76]	2025	Temporal Causality	Paradigm IV	☆	★	☆	★	☆	☆
LongVT [77]	2025	Evidence Sparsity	Paradigm II	☆	★	★	☆	☆	★
LVAgent [49]	2025	Evidence Sparsity	Paradigm I	★	☆	☆	☆	☆	☆
M3-Agent [78]	2025	Context Bottleneck	Paradigm III	☆	★	★	☆	★	★
MAGNET [42]	2025	Multimodal Ambiguity	Paradigm III	★	☆	☆	☆	☆	☆
MAViD [79]	2025	Multimodal Ambiguity	Paradigm II	☆	★	☆	☆	☆	☆
Mobile-Agent-V [23]	2025	Temporal Causality	Paradigm II	☆	★	☆	★	☆	☆
MONDAY [22]	2025	Temporal Causality	Paradigm II	☆	★	☆	★	☆	☆
Mr. Video [62]	2025	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
MSR-ViR [80]	2025	Evidence Sparsity	Paradigm II	☆	★	★	☆	★	★
PANDA [67]	2025	Temporal Causality	Paradigm IV	★	☆	☆	☆	☆	☆
ProVideLLM [33]	2025	Context Bottleneck	Paradigm IV	☆	☆	☆	☆	☆	☆
ReAgent-V [52]	2025	Evidence Sparsity	Paradigm I	☆	☆	★	☆	☆	★
ReKV [81]	2025	Context Bottleneck	Paradigm III	☆	☆	☆	☆	☆	☆
ReWatch-R1 [82]	2025	Temporal Causality	Paradigm II	☆	★	★	★	★	★
RoomTour3D [19]	2025	Temporal Causality	Paradigm II	☆	★	☆	★	☆	☆
SciEducator [83]	2025	Multimodal Ambiguity	Paradigm III	★	☆	☆	☆	☆	☆
SEASON [84]	2025	Temporal Causality	Paradigm II	★	☆	☆	☆	☆	☆
Select Less, Reason More [85]	2025	Evidence Sparsity	Paradigm I	☆	☆	★	☆	★	★
StreamBridge [86]	2025	Temporal Causality	Paradigm IV	☆	★	☆	★	☆	☆
StreamChat [31]	2025	Context Bottleneck	Paradigm III	★	☆	☆	☆	☆	☆
Thinking with Videos [87]	2025	Evidence Sparsity	Paradigm II	☆	☆	★	☆	☆	★
Time-R1 [88]	2025	Temporal Causality	Paradigm II	☆	☆	★	☆	☆	★

TABLE II
COMPARISON OF BENCHMARKS RELEVANT TO AGENTIC VIDEO UNDERSTANDING. YEAR DENOTES THE FIRST PUBLIC RELEASE OF EACH BENCHMARK. VIDEO MODE DESCRIBES ACCESS TO VIDEO EVIDENCE RATHER THAN WHETHER AN EXTERNAL WEBSITE OR TOOL IS ONLINE. EVALUATION ITEMS REPORTS THE NUMBER OF EVALUATION INSTANCES. FOR BENCHMARKS WITH TRAINING AND TEST SPLITS, REPORTED SCALE STATISTICS REFER TO THE TEST SET UNLESS OTHERWISE NOTED. DURATION VALUES ARE AVERAGES UNLESS A RANGE OR UPPER BOUND IS SHOWN. ABBREVIATIONS: MCQ = MULTIPLE-CHOICE QUESTION AND QA = QUESTION ANSWERING.

Benchmark	**Year**	**Video Mode**	**Len. (s)**	**Video Scale**	**Evaluation Items**	**Evaluation Format**	**Link**
Charades-STA [89]	2017	offline	∼30	1,334 videos	3,720	Temporal grounding	
MSRVTT-QA [106]	2017	offline	15.2	2,990 videos	72,821	Open-ended QA	
YouCook2 [91]	2017	offline	315.6	210 videos	210	Procedure segmentation	
ActivityNet-QA [90]	2019	offline	180.0	800 videos	8,000	Open-ended QA	
How2QA [107]	2020	offline	15.3	1,166 videos	2,852	MCQ	
NExT-QA [92]	2021	offline	44.0	1,000 videos	8,564	MCQ + open-ended QA	
EgoSchema [93]	2023	offline	180.0	5,063 clips	5,063	MCQ	
MVBench [94]	2023	offline	16.0	3,641 videos	4,000	MCQ	
Ego-Exo4D [108]	2023	offline	156.0	1,121 test takes	Task-dependent	Multi-task video understanding benchmark	
HiREST [109]	2023	offline	263	1,391 videos	546	Video retrieval; moment retrieval & segmentation; step captioning	
Perception Test [110]	2023	offline	23.0	3,525 videos	13,387	MCQ + grounded QA	
Video-Bench [111]	2023	offline	56.0	5,917 videos	17,036	MCQ	
LongVideoBench [95]	2024	offline	473.0	3,011 videos	5,341	MCQ	
LVBench [96]	2024	offline	4,101.0	103 videos	1,549	MCQ	
MLVU [97]	2024	offline	180.0–7,200	1,334 videos	2,593	MCQ + open-ended generation	
StreamingBench [100]	2024	online	3–1,440	900 videos	4,500	Timestamped MCQ; sequential QA; proactive output	
TempCompass [98]	2024	offline	11.4	410 videos	7,540	MCQ + binary QA + captioning	
Video-MME [99]	2024	offline	1,017.9	900 videos	2,700	MCQ	
MMBench-Video [112]	2024	offline	165.4	609 videos	1,998	Open-ended QA	
LongVALE [113]	2024	offline	235.0	1,171 videos	13,867	Temporal grounding; dense video captioning & segment captioning	
STAR [114]	2024	offline	29.7	955 videos	7,377	MCQ	
OVO-Bench [101]	2025	online	–	644 videos	3,100	MCQ + open-ended QA	
Video-MMMU [115]	2025	offline	506.2	300 videos	900	MCQ	
OmniMMI [102]	2025	online	324.3	1,121 videos	2,290	Timestamped QA + action selection	
Video-TT [116]	2025	offline	< 65	1,000 videos	5,000	Open-ended QA; MCQ; Natural-adversarial robustness	
Ego2Web [104]	2026	offline	180.0	500 pairs	500	Web-task completion	
OmniPro [103]	2026	online	189.0	1,262 videos	2,700	Proactive streaming QA; response timing	
VideoDR [105]	2026	offline	–	500	500	Open-web factoid QA; browser search	

space [9]–[11], [44], [63], [64]. Although effective for instruction following, tool use, and retrieval, this pipeline may lose fine-grained visual dynamics such as motion, state changes, interactions, and action consequences.

Video-native agent modeling instead asks whether perceptual backbones can represent these dynamics directly and expose them to downstream agent reasoning. Future agents may use representations that capture motion, state transitions, predictive scene evolution, and action-conditioned outcomes, rather than treating video models mainly as feature providers for an LLM controller. Research on world models, generative control, and video-to-action learning provides useful building blocks for this direction [20]–[22], [118]–[120]. A central question is whether such backbones improve the perception of state changes, action anticipation, and downstream task success, particularly in procedural, manipulation, and embodied settings.

VII. CONCLUSION

Video understanding agents shift video modeling from fixed inference to adaptive evidence control. This survey organizes the field through video-specific challenges, state-space paradigms, learning mechanisms, supervision, and evaluation. Future progress requires both agentic-native video modeling for operable temporal states and video-native agent modeling for reasoning over visual dynamics. Evaluation should also extend beyond final accuracy to evidence use, response timing, tool decisions, cost, and interaction reliability.

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